

Job Sequencing

Profits :- 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11

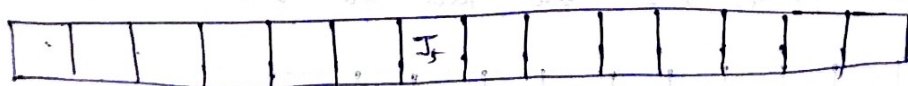
deadlines :- 3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1

step-1:- arrange profits in descending order.

<u>Job</u>	<u>Profit</u>	<u>deadline</u>
J ₅	30	7
J ₃	29	8
J ₄	28	6
J ₇	27	10
J ₁₂	27	2
J ₁₀	26	12
J ₈	25	4
J ₉	24	6
J ₁	22	3
J ₆	21	5
J ₂	19	3
J ₁₃	19	14
J ₁₁	14	18
J ₁₄	11	1

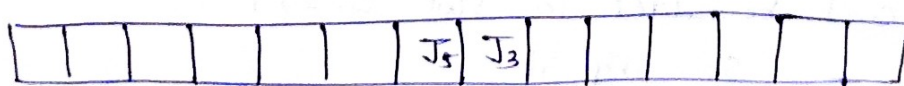
Step 2:-

(1) add J₅



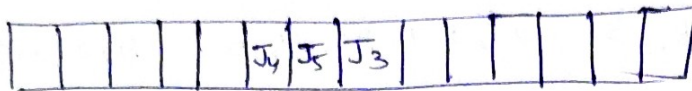
Profit = 30

(2) add J₃



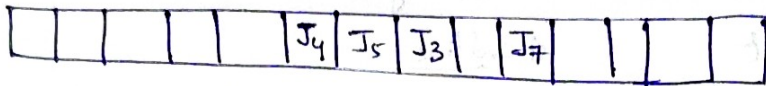
Profit = 30 + 29
= 59

add J_4



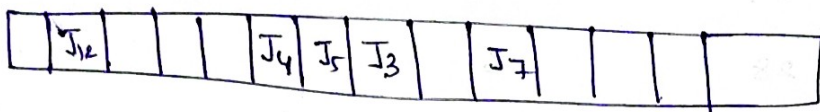
$$\text{profit} = 59 + 28 = 87$$

add J_7



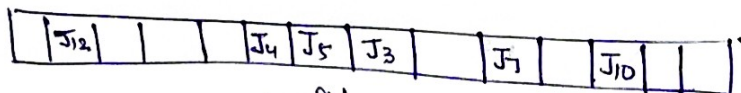
$$\text{profit} = 87 + 27 = 114$$

add J_{12}



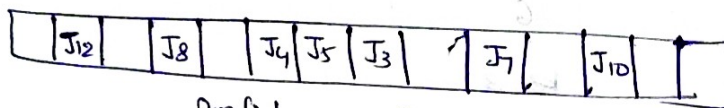
$$\text{profit} = 114 + 27 = 141$$

add J_{10}



$$\text{profit} = 141 + 26 = 167$$

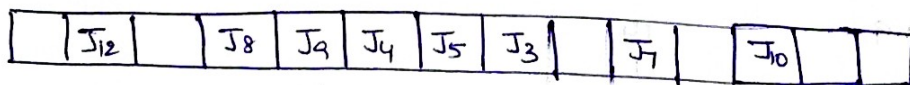
add J_8



$$\text{Profit} = 167 + 25 = 192$$

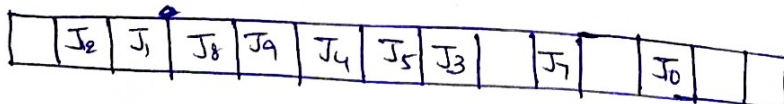
add J_9 (24, 6)

6th slot is occupied. So add to 5th as it is empty.

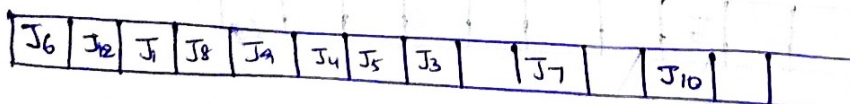


$$\text{profit} = 192 + 24 = 216$$

add J_1 (22, 3)



add J_6 slot - 5, 4, 3, 2 are full so add in slot - 1



add J_2

It can't be added in slot - 3, 2, 1
so skip it.

add J_{13}

J_6	J_2	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{13}
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$$\begin{aligned}\text{profit} &= 259 + 19 \\ &= 278\end{aligned}$$

add J_{11}

J_6	J_2	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
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$$\begin{aligned}\text{profit} &= 278 + 14 \\ &= 292\end{aligned}$$

add J_{14} (If can't be added in slot - 1)

J_6	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
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final :-

J_6	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
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